



Private LLMs with OpenSearch Agents

A field guide to running agents on a fully private LLM stack.

Abdul Muneer NetApp Instaclustr | **Rudraksh Karpe** Simplismart



SPEAKER INTRODUCTION



SPEAKER

Abdul Muneer

Open Source Developer @ NetApp Instaclustr

Maintainer: OpenSearch MLCommons

Committer: OpenJDK

linkedin.com/in/muneer-kolarkunnu/ github.com/akolarkunnu

SPEAKER INTRODUCTION



SPEAKER

Rudraksh Karpe

Forward Deployed Engineer · Simplismart

Builds solutions solutions focused on high-performance AI inference, GSoC Mentor
@openSUSE

Previously presented at conferences such as PyTorch Conf (US & EU), Kubecons EU, PyCon(US & JP), OpenSearchCon, etc. across the globe.

rudrakshkarpe.com [linkedin.com/rudraskhkarpe](https://www.linkedin.com/in/rudraskhkarpe) x.com/rudrakshkarpe github.com/rudrakshkarpe

What this talk covers

01

The landscape

Why private LLMs, and what OpenSearch Agents are.

02

Agents & tools

Agent types and how agentic search really works.

03

Connecting a private LLM

Connectors, the trusted-endpoint wall, and the setup path.

04

Certificate hell + fixes

JDK keystore, disabling validation, and our PRs.

05

End-to-end & the new UI

Full private architecture and the 3.4 Agentic UI.

06

Live demo + takeaways

Production and local, then what to remember.

Why run LLMs privately?



Data sovereignty

Sensitive data never leaves your network.



Air-gapped

Runs where public APIs can't reach.



Compliance

GDPR, HIPAA, RBI, no third-party sharing.



Cost & control

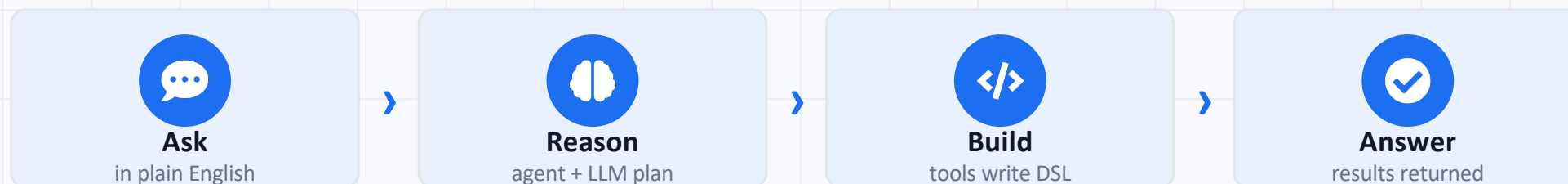
No per-token bill. You own the lifecycle.



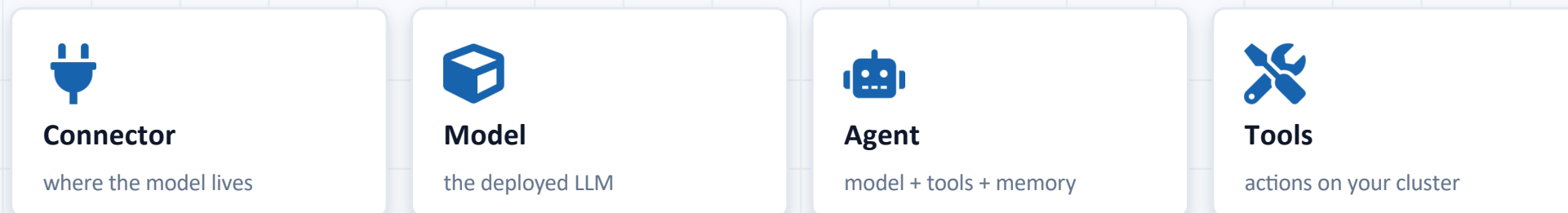
The catch OpenSearch assumes a public endpoint over valid TLS. Private LLMs break that, here's how we fixed it.

What are OpenSearch Agents?

An agent lives in ML Commons. It uses an LLM to reason, pick tools, and turn a plain-English question into a real OpenSearch query.



FOUR BUILDING BLOCKS



Four agent types, pick by job



Flow

Fixed tool sequence. Fast, cheap, no memory.

[QUICK NL → DSL](#)



Conversational

Multi-tool reasoning with memory for follow-ups.

[MULTI-TURN CHAT](#)



Plan-Execute-Reflect

Plans, runs, and refines multi-step work.

[RESEARCH & RCA](#)



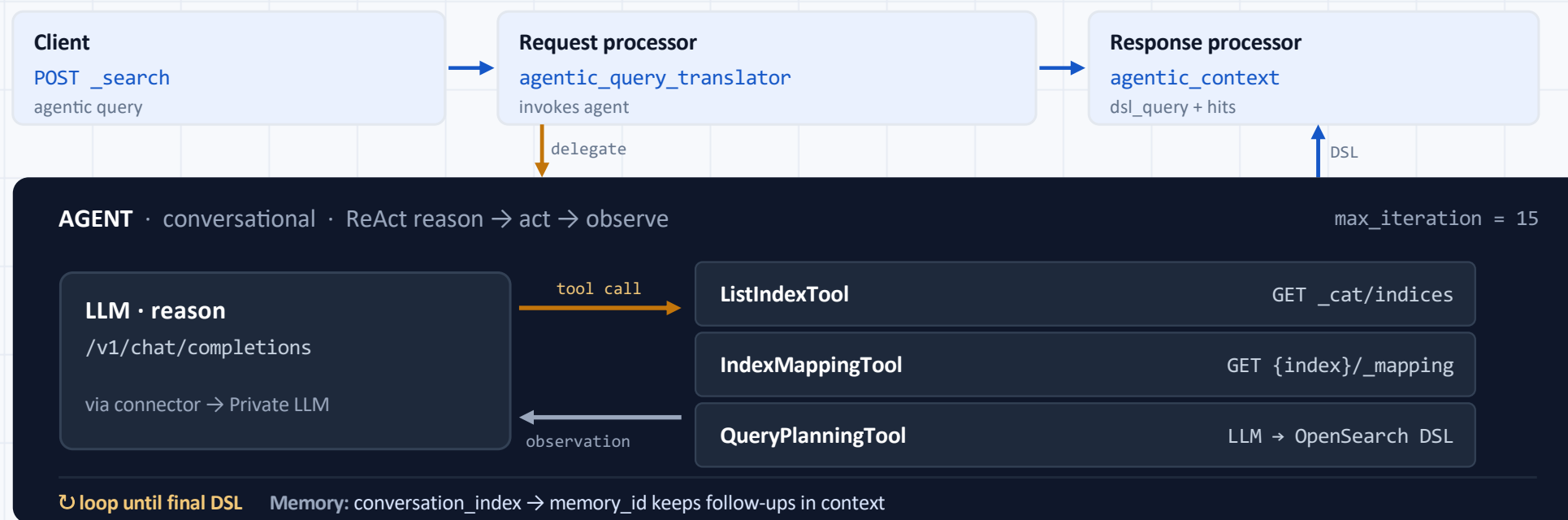
AG-UI

Drives interactive frontend components live.

[INTERACTIVE UI](#)

Rule of thumb start with Flow; reach for Plan-Execute-Reflect only when the question needs real multi-step reasoning.

How agentic search works



Output → search hits + `agent_steps_summary` (full execution trace)

Our journey: connecting a private LLM

It all worked with a hosted model. Then we pointed it at our own LLM, and hit three walls.

The path to a working agent

On paper it's five REST calls against ML Commons. Clean, until step 1 meets a private model.



Steps 1–3 are where a private LLM falls off the happy path. Three walls, coming up.

The trusted-endpoint guard

THE SYMPTOM

```
POST .../_register  
→ illegal_argument_exception - Connector  
URL is not matching the trusted connector  
endpoint regex
```

```
POST .../_predict  
→ illegal_argument_exception - Remote  
inference host name has private ip  
address
```

ML Commons only allows endpoint URLs that match a regex, and blocks private ips. A local model fails instantly.

THE FIX, TWO SETTINGS

```
PUT /_cluster/settings  
{ "persistent" : {  
  
  "plugins.ml_commons.connector.private_ip_enabled": true,  
  
  "plugins.ml_commons.trusted_connector_endpoints_regex": [  
    "^https://private-ip/.*$"  ]  
}
```

private_ip_enabled; the regex whitelists your host. Keep it narrow in prod.

Certificate hell

Our private LLM sat behind a self-signed / corporate CA. OpenSearch runs on the JVM, which trusts nothing it can't verify.

```
POST .../_predict
javax.net.ssl.
SSLHandshakeException:
  PKIX path building failed -
  unable to find valid
  certification path to target
```

WHY IT HAPPENS

- Cert signed by a private CA
- CA isn't in the JVM truststore (cacerts)
- Handshake fails, connector never connects

TWO ROADS OUT



A • Trust the cert

Import the CA into the keystore. Production-grade.



B • Skip validation

Disable checks per-connector. Local / test only.

Trust the CA in the keystore

Add the CA to the JVM truststore (cacerts).

```
# Import into the truststore
keytool -importcert -alias private-llm \
  -file llm-ca.pem \
  -keystore "$JAVA_HOME/.../cacerts" \
  -storepass changeit -noprompt
```

A per-connector escape hatch

For local & test, importing certs everywhere is heavy. There was no per-connector "skip TLS" flag, so we filed it and proposed the option.



Disable cert validation for LLM connectors



Move disable cert parameter to client config



Self-signed handling in HTTP client

```
POST /_plugins/_ml/connectors/_create
{
  "name": "Private LLM connector",
  "protocol": "http",
  "client_config": {
    "skip_ssl_verification": "true"
  },
  ...
}
```



USE RESPONSIBLY

Local & test only, never prod. An issue + PR beats a private workaround.

An Ollama blueprint

OpenSearch ships blueprints for hosted providers, but none for Ollama, the easiest private model to run. So we wrote one.

```
POST .../connectors/_create
{
  "name": "Ollama Chat",
  "protocol": "http",
  "parameters": {
    "model": "llama3.1"
  },
  "actions": [{
    "url":
      "http://localhost:11434/..."
  }]
}
```



Ollama blueprint

Template for local models



Cert-skip option

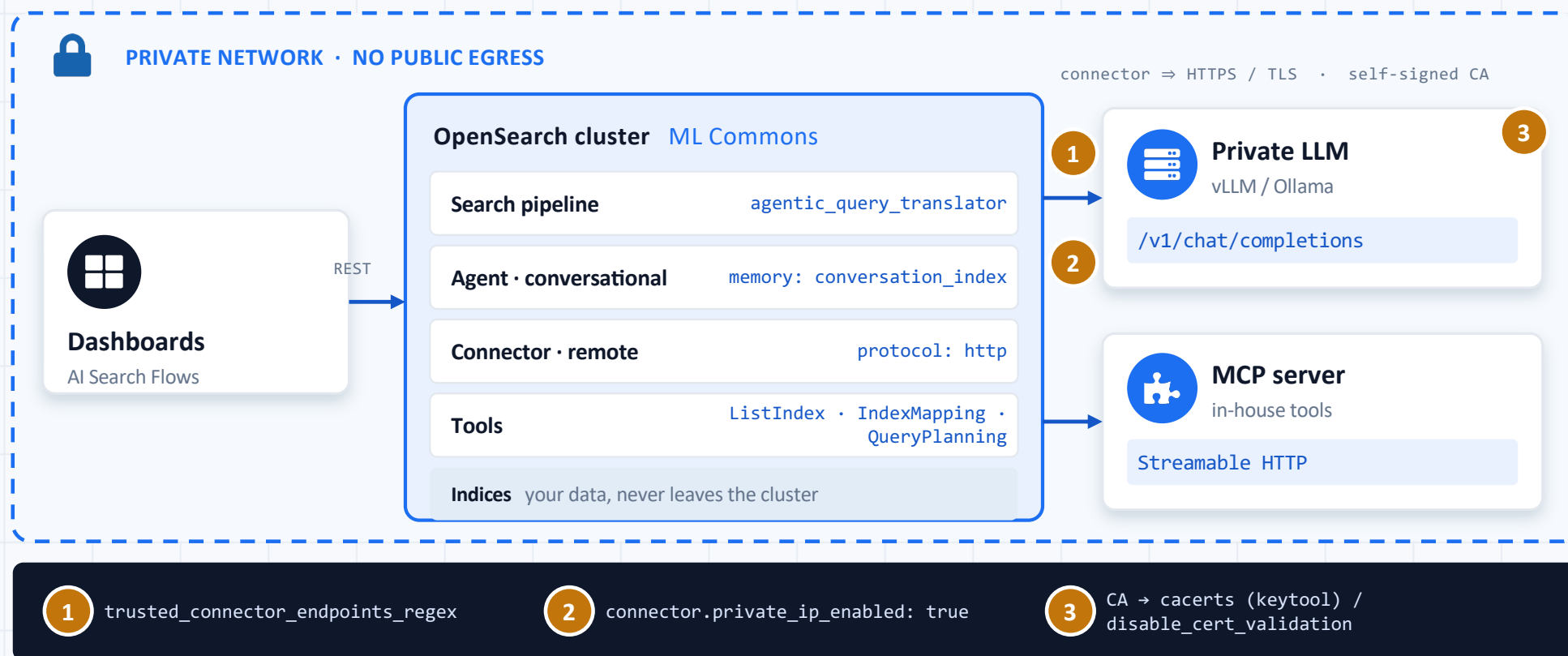
Built in for local testing



Private-IP guidance

Keys for loopback hosts

The private setup, nothing leaves



The new Agentic UI

A no-code experience: Plugins → AI Search Flows. Build an agent and run agentic search straight from Dashboards.

AGENT CONFIG

Agent type · Model
+ QueryPlanningTool
+ ListIndexTool
+ IndexMappingTool
External MCP · Memory

SEARCH EXECUTION

"Which orders shipped late?"

```
✓ ListIndexTool → orders  
✓ IndexMappingTool → schema  
✓ QueryPlanningTool → DSL  
{ "bool": { "filter":  
  ship_date > sla } }
```

No code

Build & test without hand-writing JSON

Private LLMs with OpenSearch Agents

MCP + templates

Attach external MCP servers in the UI

Memory & traces

Multi-turn chat with step-by-step reasoning

The new Agentic UI

local-ollama-agentic-search-demo ⚙️

Last saved: 06/14/26 06:57 AM Close Export

Configure agent EXPERIMENTAL ⓘ

Agent Local Ollama Prompted Conversational Agent ▾

Form

JSON

+ Create new agent

Name

Local Ollama Prompted Conversational Agent

Type

Conversational 🔒

Description

Prompted agentic search demo agent backed by local Ollama llama3.1

Model

Local Ollama llama3.1 agent model ▾

Learn more ↗️

Tools

> Query Planning - required 🔴 Enable

> Search Index 🔴 Enable

> List Index 🔴 Enable

> Index Mapping 🔴 Enable

> Web Search 🔴 Enable

MCP servers

+ Add MCP server

Advanced settings

Learn more ↗️

Agentic search

Index agentic-products-demo-1781399217 ▾ 🔍

Agent Local Ollama Prompted Conversational ... ▾ 🔍

🔍 Query ⓘ

Continue conversation ⓘ

Form

JSON

🔍 Search

Show me in-stock Adidas running shoes under 200 dollars

+ Add query fields ⓘ

⚙️ Generated query ⓘ

👁️

```
{
  "query": {
    "bool": {
      "filter": [
        {
          "term": {
            "category": "shoes"
          }
        },
        {
          "range": {
            "price": {
              "lt": 200
            }
          }
        }
      ]
    }
  }
}
```

Model Registration command

POST /_plugins/_ml/models/_register

```
{
  "name": "OpenSearchCon India Test Model",
  "function_name": "remote",
  "description": "test model",
  "connector": {
    "name": "OpenSearchCon India Test Connector",
    "description": "The connector to openai chat model",
    "version": 1,
    "protocol": "http",
    "client_config": {
      "skip_ssl_verification": "true"
    },
    "parameters": {
      "model": "gpt-5"
    },
    "credential": {
      "openAI_key": "your_key"
    },
    "actions": [
      {
        "action_type": "predict",
        "method": "POST",
        "url": "https://private_ip",
        "headers": {
          "Authorization": "Bearer ${credential.openAI_key}"
        },
        "request_body": "{ \"model\": \"${parameters.model}\", \"messages\": [
          [ { \"role\": \"developer\", \"content\": \"${parameters.system_prompt}\", \"${parameters._chat_history:-\"\"} \"${parameters.user_prompt}\" ], [ { \"role\": \"user\", \"content\": \"${parameters.interaction}\" } ] ] }"
      }
    ]
  }
}
```



LIVE DEMO

Private LLMs, two ways



Production A hardened private LLM behind a trusted CA → a conversational agent.



Local / testing Ollama on a laptop → the Dashboards Agentic UI.

Two paths, side by side



In production

Hardened, fully private

- 1 CA already imported into cacerts on every node
- 2 Register connector over HTTPS → model → conversational agent
- 3 Build the agentic search pipeline
- 4 Ask: "Top 5 error-prone services this week?"



Local / testing

Laptop in minutes

- 1 ollama run open model on localhost
- 2 Set private_ip_enabled + endpoint regex
- 3 Connector with disable_certificate_validation
- 4 Dashboards → AI Search Flows → ask away

Key takeaways

1

Private is possible

On-prem agents work today, the hurdles are endpoints, IPs and certs.

2

Certificates are the crux

Trust the CA in prod; skip validation only for local testing.

3

The community fills gaps

Missing a blueprint or a flag? File an issue, open a PR.

4

Choose the right agent

Flow for quick NL→DSL; reflect only for real multi-step work.



Q & A

Ask us anything, walls you've hit, certs, MCP, or the demo.

Thank you!

Go build something private. We can't wait to see it.

